

REVIEW

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A comprehensive review of machine learning applications in forecasting solar PV and wind turbine power output

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Abstract

With climate change driving the global push toward sustainable energy, the reliability of power systems increasingly depends on accurate forecasting methods. This study examined the role of machine learning (ML) in forecasting solar PV power output (SPVPO) and wind turbine power output (WTPO) and identified the challenges posed by the intermittent nature of these renewable energy sources. This study examined the current techniques, challenges, and future directions in ML-based forecasting of SPVPO and WTPO and proposed a standardized framework. Using the Mann–Whitney and Kruskal–Wallis tests, the results highlight the significant impact of key meteorological and operational variables on enhancing forecasting accuracy, as measured by MAPE and R-squared. Key features for SPVPO forecasting include solar irradiance, ambient temperature, and prior SPVPO, while wind speed, turbine speed, and prior wind power output are crucial for WTPO forecasting. Moreover, ensemble models, support vector machines, Gaussian processes, hybrid artificial neural networks, and decomposition-based hybrid models exhibit promising forecasting accuracy and reliability. Challenges such as data availability, complexity-interpretability trade-offs, and integration difficulties with energy management systems present opportunities for innovative solutions. These include exploring advanced data processing and calibration techniques, leveraging Big Data and IoT advancements, formulating advanced machine learning (ML) techniques, and employing probabilistic approaches with desirable accuracy and robustness in forecasting solar photovoltaic power output (SPVPO) and wind turbine power output (WTPO). Additionally, expanding research to ensure model generalizability across diverse climate conditions and forecasting horizons is crucial for enhancing the reliability and efficiency of renewable energy forecasting using machine learning techniques.

Keywords: Solar PV power output forecasting, Wind turbine power output forecasting, Forecasting, Core principles and forecasting accuracy, Machine learning-based forecasting techniques, Solar and energy forecasting models

Introduction

The global movement for sustainable energy sources, driven by climate change, has promoted the increase in renewable energy utilization, such as solar PV and wind turbines. However, the intermittent nature of these sources poses challenges to power system

reliability [1, 2], requiring accurate forecasting to address issues arising from weather conditions and seasonal changes. Several studies emphasize the importance of accurate power generation forecasting when employing solar and wind power to address climate change [3–5]. Future directions in forecasting renewable energy, including the application of deep learning methods and the necessity for probabilistic information, are discussed in [6] and [7].

Accurate forecasting plays a crucial role in energy system management and the integration of solar and wind energy into the grid. It helps with grid management, reduces the need for backup power, and stabilizes energy infrastructure. Various forecasting models have been developed in recent years to produce accurate forecasts, including statistical methods, artificial intelligence (AI), machine learning (ML), and hybrid approaches. Essentially, ML techniques have enhanced the accuracy and efficiency of energy forecasting processes, particularly in load forecasting, renewable energy integration, demand response optimization, and asset maintenance and optimization [8–10]. Based on the context and purpose, forecasting is generally classified depending on the time horizon. In solar PV power output (SPVPO) and wind turbine power output (WTPO) generation, the forecasting time horizon employed is crucial for efficient grid management, energy trading, and the integration of renewable energy. Studies like [11] and [12] explained the differences in forecasting time horizons in this aspect. Ultra-short-term forecasts, which range from 1 min to an hour, are primarily employed for power smoothing, real-time electricity dispatch, and optimizing energy reserves, whereas short-term forecasts, which extend from an hour to a week, are used for power plant unit commitment, scheduling dispatch, and enhancing the security of grid operations. Medium-term forecasting spans a period of 1 week to a month and is primarily used for power system planning and scheduling maintenance. On the other hand, long-term forecasting extends from 1 month to a year and is commonly applied in planning electricity generation, transmission, and distribution organizations.

Machine learning (ML) is a transformative technology that has significantly impacted the energy industry. It involves the creation of models and algorithms that let computers carry out operations that have historically required human intelligence, such as problem-solving, decision-making, and pattern recognition [8, 9]. Among the standard ML techniques for forecasting solar and wind energy production, artificial neural networks (ANNs) are popular due to their ability to learn complex behaviors and nonlinear patterns in historical data and meteorological factors [13, 14]. ANNs have succeeded in various applications, particularly in forecasting SPVPO and WTPO. ANNs include multilayer perceptron (MLP), convolutional neural networks (CNNs), and recurrent neural networks (RNNs). Specifically, long short-term memory (LSTM) networks are enhanced recurrent neural networks (RNNs) well-suited for sequence modeling and time series forecasting [15]. They can resolve the vanishing gradient issue of RNNs and their limitations by learning long-range dependencies in the data. Additionally, support vector machines (SVMs) are powerful supervised learning models used for forecasting, encompassing both classification and regression. SVMs are effective in forecasting using low- or high-dimensional data, as well as in cases where class discrepancies exist [16].

Moreover, random forests (RFs) are ensemble learning techniques for both classification and regression-based forecasting, aggregating the results of multiple decision trees

to resolve potential overfitting [17–19]. RFs are robust enough to model overfitting and effectively deal with high-dimensional data, a common challenge in energy forecasting problems. Similarly, gradient boosting machines (GBMs) are ensemble techniques that improve predictive power by training iteratively [20]. They build strong predictive models by introducing multiple decision trees and correcting errors at each step. GBMs are powerful in renewable energy forecasting because they can learn from complex data without compromising accuracy.

Furthermore, Gaussian processes (GPs) are nonparametric probabilistic models that utilize the Gaussian distribution of input and output variables, providing a flexible framework for modeling complex relationships in data while quantifying uncertainty estimates [21–23]. GPs are particularly effective for cases where data exhibit nonlinear and non-stationary patterns, such as SPVPO and WTPO forecasting. Lastly, hybrid models (HMs) combine multiple forecasting techniques or incorporate domain knowledge and expert insights to improve forecasting accuracy. They integrate forecasts generated by different individual models, such as ML and statistical models, to capture various aspects of the underlying patterns in the data. This can enhance the performance and interpretability of SPVPO and WTPO forecasting systems.

To increase the precision of SPVPO forecasting, there have also been suggestions for integrating crucial meteorological variables extracted from photovoltaic models [24]. Factors such as environmental conditions, dust, and technological impacts can affect the performance of solar panels [11, 25, 26]. Other factors include the type of PV, orientation, parasitic resistances, cable thickness, clouds, inverter efficiency, dust accumulation, PV orientation, weather conditions, and location [27–30]. In forecasting SPVPO, a comparison between the use of different types of solar irradiance [31, 32] and the use of solar irradiance combined with other weather parameters was investigated [33]. Studies like [34] and [35] identified ambient temperature (AT), relative humidity (RH), visibility, air pressure (AP), wind speed (WS), cloud cover (CC), wind bearing (WB), and precipitation (TP) as key meteorological variables in forecasting SPVPO. Similarly, [36] emphasizes the role of numerical weather prediction (NWP) data, including SI, TP, WS, and AT. The impact of soiling on PV plant performance is also a crucial consideration, as discussed by [37].

On the other hand, WTPO is produced by harnessing kinetic energy through wind turbines, which have undergone significant evolution due to research and development efforts [38]. Land utilization, weather data, and configuration of wind turbines are key factors in calculating wind power potentials [39–42]. Additionally, weather variables in WTPO forecasting have significantly improved the predictability and management of climate variability-related risks [39]. Seasonal climate predictions of wind speed have proven useful in the wind power industry [40]. The stochastic nature of wind power generation is addressed, with a focus on the state-of-the-art in energy forecasting [41]. Additionally, the impact of weather parameters, particularly air density influenced by pressure, humidity, and temperature, on WTPO is discussed [42].

These findings provide important insights crucial for improving forecasting models in the energy sector. The literature on AI and ML technologies in energy forecasting is limited, as most studies focus on their more general applications. This requires continued research to obtain optimal machine learning techniques and models in the domain.

The use of big data and deep learning models has been particularly effective in improving the accuracy of energy forecasting, with deep neural network architectures and hybrid methods showing promising results [43–45]. Selecting the most suitable technique depends on various factors such as data size, variability, and environmental variables. Table 1 summarizes the existing academic review literature, focusing on the results and outcomes of each study, covering various topics such as the categorization of prediction techniques, the performance of different machine learning models, and the obstacles and advancements in forecasting SPVPO and WTPO.

It is evident in the literature that there has been a lack of standardized methodology and approach for SPVPO and WTPO forecasting, which increases the complexity of forecasting operations. This encompasses data processing, feature selection and evaluation, and ML models. For instance, despite the recent advancement in solar energy forecasting research, [56] highlighted the necessity to standardize the forecasting practice for solar power. With this, the Reproducible, Operational, Probabilistic, or Physically Based Ensemble and Skill (ROPES) guideline was introduced to resolve this ongoing concern, allowing effective and comprehensive model development, comparison, and integration for solar energy forecasting. A standard methodology is also needed to facilitate easier decision-making for grid operators and energy planners, allowing for the seamless integration of SPVPO and WTPO into the global energy infrastructure. There is also a lack of comprehensive analysis of forecasting procedures and performance for SPVPO and WTPO forecasting. However, the existing studies highlight promising developments in this domain. Several studies have demonstrated the potential of machine learning-based forecasting methodologies tailored explicitly for SPVPO and WTPO predictions [57].

Table 1 Summary of related review literature published

Source	Scope
[46]	They surveyed AI methods for renewable energy forecasting, discussing their taxonomy for solar and wind energy, combining ML and metaheuristic methods, and presenting potential improvements
[47]	Discusses current methods in PV power prediction, emphasizing the use of ML and deep learning, particularly ANNs for short-term forecasts, and the need for hardware implementations for real-time predictions
[48]	It classifies PV power forecasting based on the ML family, location, climate, system count, period, forecast horizon, and feature selection
[49]	Identifies parameters affecting AI-based SPVPO forecasting, explores model types and forecast time steps, and suggests techniques to improve accuracy and stability
[50]	Summarizes current trends of ML-based models for SPVPO and WTPO forecasting in terms of the forecast horizon, data collection and preprocessing techniques, feature extraction and selection, hyperparameter optimization, and forecast accuracy
[51]	Reviews ML applications for SPVPO forecasting across time horizons, assessing model effectiveness and identifying methods to enhance forecast accuracy and reliability
[52]	Reviews ML applications in offshore wind energy, addressing development, installation, operation, and maintenance challenges
[53]	Critically reviews ML methods for PV power, highlighting the frequent use of ANNs and SVMs
[54]	Discusses the effectiveness of LSTM and DCNN for short-term SPVPO forecasting, emphasizing the importance of optimizing input data and standardizing models
[55]	It reviews metaheuristic and ML methods for forecasting SPVPO, compares their pros and cons, and provides a comprehensive comparison for future research
This study	Provides a comprehensive review of ML forecasting of SPVPO and WTPO and a statistical comparison of forecasting accuracy across various models and techniques

Moving forward, a common framework for forecasting SPVPO and WTPO must be used. The main objective of this study is to thoroughly evaluate and enhance the domain of SPVPO and WTPO forecasting by examining the fundamental concepts, methodology, and incorporation of ML. Specifically, this study aimed to (1) investigate the fundamental principles and variables influencing SPVPO and WTPO generation, assessing their influence on forecasting accuracy; (2) identify SPVPO and WTPO machine learning forecasting methods, evaluate the limitations of these methods, and highlight their potential to enhance accuracy and efficiency; and, (3) identify the challenges and opportunities in integrating ML for SPVPO and WTPO forecasting. This provides a valuable synthesis of the literature on recent trends, challenges, and future directions in ML applications for SPVPO and WTPO forecasting.

The paper is organized as follows to facilitate reader navigation: Sect. “[Methodology](#)” covers the process for selecting, categorizing, and evaluating the relevant literature. A thorough summary of the machine learning models used for SPVPO and WTPO forecasting is provided in Sect. “[Core principles and forecasting accuracy](#)”, along with a comparison of their capabilities, drawbacks, and strengths. The current issues and new prospects in this field are examined in Sect. “[Machine learning-based forecasting techniques](#)”. Sect. “[Challenges and opportunities](#)” concludes the work by highlighting key discoveries and providing suggestions for future research avenues.

Methodology

Literature selection process

This study employed the Systematic Review Process detailed by [58, 59], and [60] to select literature corresponding to SPVPO and WTPO forecasting. The initial screening process consisted of three stages: establishing relevant keywords, creating a Boolean string search for Advanced Search in the Scopus database, and applying additional filters to recent English-language literature from 2019 to 2023. The study employs the following Boolean string search method for the selected keywords:

- Solar AND PV AND output AND forecasting AND using AND machine AND learning
- Wind AND power AND output AND forecasting AND using AND machine AND learning

This resulted in 94 literatures for SPVPO and 151 for WTPO forecasting. The final screening stage then involved manual evaluation of the alignment of each literature with the research objective, title, abstract, keywords, introduction, methodology, and conclusion chapters. The study excluded literature on hybrid solar-wind energy production, resulting in 74 literature for SPVPO and 57 for WTPO forecasting. Consequently, the results of this study depend on the quality of the data available on the Scopus database. The topic’s broad nature may pose challenges in comprehensively covering all relevant aspects.

Evaluation criteria

This study evaluated the literature on SPVPO and WTPO forecasting using ML methods, data quality and processing techniques, meteorological factors, forecasting accuracy metrics, model evaluation results, and limitations and challenges encountered. The process involved systematically organizing and summarizing results to provide insights into the performance, effectiveness, limitations, and challenges of different models and techniques. Techniques employed included evaluating the forecasting time horizon, data quality, and data processing; characterizing machine learning models; emphasizing novel approaches; identifying common trends and patterns across studies; influencing variable selection and preprocessing techniques; examining trade-offs between model complexity and accuracy; and discussing potential biases or limitations. The study also compared the performance of machine learning models against traditional statistical methods or physics-based models. The findings provide valuable insights into advancements and challenges in the field.

Data and tools

To gain an overview of the selected relevant literature on SPVPO and WTPO forecasting, the literature was summarized according to its publication year and the country or countries under study. Microsoft Excel and R version 4.3.2 were used for data consolidation and statistical analysis, respectively.

In terms of publication year, over half of the retrieved literature on SPVPO and WTPO forecasting was published in the last 2 years alone. Table 2 shows the distribution of retrieved literature based on the publication year in Scopus.

Furthermore, the top countries under study for SPVPO forecasting are India and Australia. Meanwhile, in WTPO forecasting, most of the literature was for China. Figures 1 and 2 show the distribution of retrieved literature based on the country or countries under study for SPVPO and WTPO forecasting, respectively.

Lastly, regarding the forecasting horizons, the retrieved literature concentrated on ultra-short-term (27.9%) and short-term (55.8%) horizons. Hence, evaluating and analyzing the ML models and factors were relevant primarily to ultra-short- and short-term forecasting.

Table 2 Temporal distribution of retrieved literatures based on publication year

Publication year	Solar		Wind	
	Count	% to total	Count	% to total
2019	9	12.2%	7	12.3%
2020	9	12.2%	7	12.3%
2021	14	18.9%	14	24.6%
2022	19	25.7%	13	22.8%
2023	23	31.1%	16	28.1%
Total	74	100.0%	57	100.0%

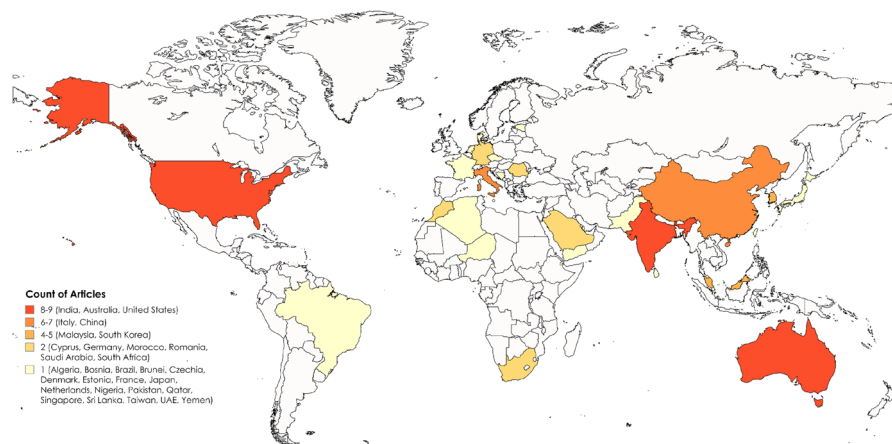


Fig. 1 Spatial distribution of retrieved literature for SPVPO forecasting

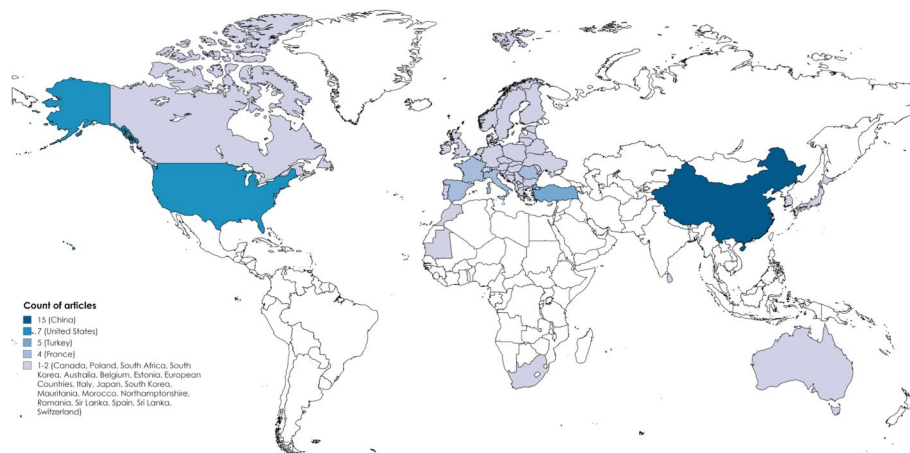


Fig. 2 Spatial distribution of retrieved literature for WTPO forecasting

Evaluation of features and ML models

The meteorological variables, solar PV characteristics, and wind turbine characteristics were then evaluated based on their data sources, collection processes, data periods, data resolutions, and impacts on forecasting accuracy. The study aimed to explain how these factors influence SPVPO and WTPO production and their impact on forecasting accuracy. It also compared the forecasting accuracy of the SPVPO and WTPO forecasting models, determining the metrics widely used in the reviewed studies. Comparisons of ML techniques and models were conducted by examining the differences in forecasting accuracy measures across various ML-based approaches.

Statistical parametric tests, such as *t* tests and analysis of variance (ANOVA), can be used to test and compare each error metric across different models. However, factors such as the ML model used, data quality, forecasting horizon, and geographical and temporal differences can vary the forecasting accuracy across different groups, thereby violating the statistical assumptions on normality and equality of variance [61]. Nonparametric counterparts of the *t* test and ANOVA—Mann–Whitney U test and

Kruskal–Wallis H test, respectively, were employed. The significance level for both tests was set at 0.05, indicating statistical evidence that the median forecasting accuracy significantly differs between or across groups of models under comparison. Post hoc analyses through pairwise comparisons through the Mann–Whitney test were performed for three or more groups of models. Unitless error metrics, such as MAPE and R-squared, were used for model comparisons. MAPE indicates the average of the absolute percentage errors committed by the forecasting models, where a lower value suggests higher forecast accuracy. Meanwhile, R-squared reflects the proportion of variability in the historical data that the forecasting model explains; thus, a higher value indicates greater model accuracy. It is important to note that the inferences from these tests do not imply causality, as they only test differences in forecasting accuracy where an important feature is included.

Synthesizing findings from diverse ML forecasting studies is crucial for understanding the research landscape and identifying best practices. By examining the impact of feature inclusion on model performance and analyzing the differences in forecasting accuracy measures across various ML-based approaches, this study contributes to the understanding of the landscape of research in the field of ML forecasting.

Core principles and forecasting accuracy

The standard features considered candidate input variables in forecasting SPVPO and WTPO were identified and ranked according to the frequency of use in the reviewed literature. Figure 3 shows the distribution of retrieved literature for SPVPO and WTPO per identified feature.

In SPVPO forecasting, the highly prominent features considered in the reviewed studies are SI (71.6%) and AT (66.2%), followed by SP (50.0%), prior power output (PPO) (48.6%), and RH (40.5%). Other fair-use features include solar PV characteristics, CC, spatial and temporal features, TP, wind direction (WD), and AP. Several solar PV energy studies have considered solar angles (SA), sunshine duration (SD), heat flux (HF), weather type (WT), and losses (L). These features are highly focused on meteorological conditions, and only a few are related to solar PV characteristics and conditions.

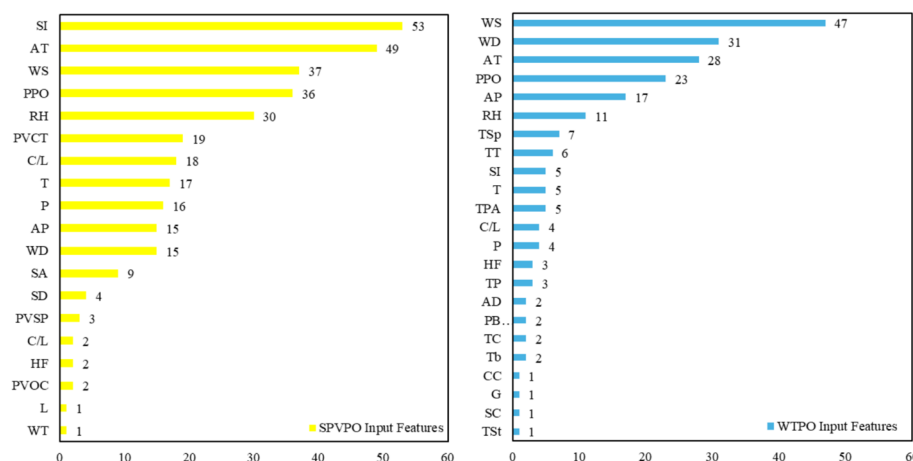


Fig. 3 Summary of features considered in SPVPO (A) and WTPO (B) forecasting

In WTPO forecasting, on the other hand, the majority (82.5%) of the studies considered WS as an essential input in the model, followed by WD (54.4%) and AT (49.1%). PPO was also commonly considered (40.4%) when forecasting future WTPO. Furthermore, it can be observed that turbine operating characteristics such as turbine (i.e., rotor and generator) speed (TSp) (12.3%), temperature (TT) (10.5%), and position and angles (TPA) (8.77%) were also reasonably considered for WTPO forecasting. A few studies also regarded spatial and temporal variables, SI, TP, HF, turbulence (Tb), PBL height, air density (AD), soiling, geopotential (G), CC, and other turbine-related variables such as operating status. SPVPO and WTPO forecasting studies seem to have a standard set of input variables; however, they differ in distribution due to the discrepancies in the objectives and interests of the researchers.

Before analyzing the potential impact of including these features on model performance, the commonly used error metrics in the reviewed studies for SPVPO and WTPO forecasting were first identified. Figure 4 shows the distribution of retrieved literature across the accuracy metrics identified for SPVPO and WTPO forecasting. RMSE and MAE are the top accuracy metrics used for both SPVPO and WTPO forecasting, followed by R-squared, MAPE, and normalized RMSE. In the SPVPO forecasting studies, other metrics include MRE, skill score, actual-forecast correlation, RMSPE, normalized MAE, and bias. Meanwhile, other metrics for WTPO forecasting are normalized MAE, bias, and interval-related metrics, particularly PICP and ACE.

The metrics for SPVPO and WTPO forecasting are highly focused on RMSE and MAE, which are scale-dependent. Among the top accuracy metrics, MAPE, R-squared, and nRMSE are unitless, making model comparisons feasible across different datasets.

To analyze the impact of the top features in the forecasting model, a set of Mann–Whitney U tests was conducted to examine and test the equality of error metrics between models with and without the top features identified. Tables 3 and 4 show the statistical comparison results of SPVPO forecasting models with and without the predictors mentioned above in terms of forecasting accuracy, particularly MAPE and R-squared. These tables consist of the following columns: *Model*, which indicates the group of ML models in terms of inclusion of the identified features for SPVPO; *Count*,

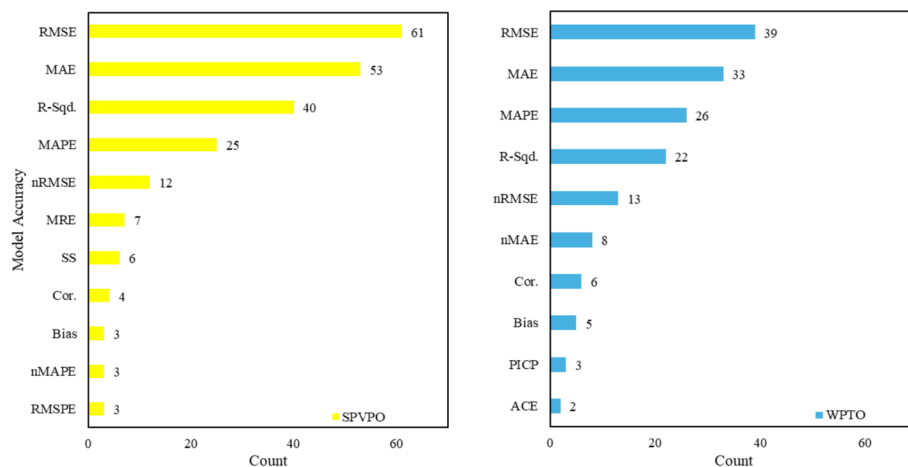


Fig. 4 Summary of accuracy metrics used in SPVPO (A) and WTPO (B) forecasting

Table 3 Statistical Comparison of MAPE and R-squared according to Inclusion of Important Features in Forecasting SPVPO using Mann–Whitney U Test

Model	MAPE			R-Squared		
	Count	Median	p value	Count	Median	p value
<i>SI</i>						
With SI	182	0.0743	0.001	312	0.9350	< 0.001
Without SI	47	0.2188		95	0.8684	
<i>AT</i>						
With AT	150	0.1408	0.019	260	0.9534	0.002
Without AT	79	0.1692		147	0.9197	
<i>WS</i>						
With WS	125	0.1301	0.536	216	0.9109	0.045
Without WS	104	0.1280		191	0.8981	
<i>PPO</i>						
With PPO	79	0.1202	0.032	160	0.9163	0.039
Without PPO	150	0.1962		247	0.8742	
<i>RH</i>						
With RH	85	0.1511	0.411	114	0.9011	0.022
Without RH	144	0.1609		263	0.8638	

Table 4 Statistical comparison of MAPE and R-squared according to inclusion of important features in forecasting WTPO using Mann–Whitney U Test

Model	MAPE			R-Squared		
	Count	Median	p value	Count	Median	p value
<i>WS</i>						
With WS	159	0.0783	0.982	175	0.9670	0.005
Without WS	49	0.0856		26	0.9130	
<i>WD</i>						
With WD	142	0.0833	0.128	117	0.9064	0.413
Without WD	66	0.0855		84	0.8765	
<i>PPO</i>						
With PPO	51	0.0964	0.317	163	0.9731	< 0.001
Without PPO	157	0.0827		38	0.9130	
<i>AT</i>						
With AT	149	0.0811	0.019	127	0.9500	0.738
Without AT	59	0.1300		74	0.9696	
<i>TS</i>						
With TS	9	0.1315	0.122	38	0.9650	< 0.001
Without TS	199	0.3315		163	0.8771	

which shows the number of ML models for SPVPO forecasting falling under the corresponding model category or group; *Median*, which indicates the median forecasting accuracy of a corresponding group of ML models in terms of MAPE and R-squared; and *p value*, which reports the corresponding p value of the Mann–Whitney U test employed in the model comparison.

Based on the results, models with SI features report a lower mean median MAPE (7.43%) than those without SI, which show a relatively higher median MAPE (21.88%). The Mann–Whitney test results provide statistical evidence of a statistically significant difference in the median MAPE values of ML models when SI is included ($p < 0.001$). This suggests that including SI may significantly improve the forecasting accuracy of the model. This underscores the critical role of solar irradiance in accurate SPVPO forecasting.

Similarly, models that include AT exhibit better forecasting accuracy, as indicated by a lower median MAPE (14.08%) compared to those without AT features, which have a median MAPE of 14.08%. The result of the nonparametric test also provides statistical evidence of significant differences in the median MAPE values between models with AT and those without ($p = 0.019$). This highlights the importance of ambient temperature as a predictive feature in SPVPO forecasting.

Conversely, including WS and/or RH may not significantly improve the forecasting accuracy, with p values of 0.536 and 0.411, respectively. The median MAPE values of models with WS (13.01%) and with RH (15.11%) are at par with those without WS (12.80%) and RH (16.09%). On the other hand, PPO shows statistical significance and may contribute to model performance improvement when included ($p = 0.032$), indicating varying degrees of impact on forecasting accuracy.

Based on MAPE values, these results suggest that while some features, such as solar irradiance, ambient temperature, and prior SPVPO, are crucial for improving accuracy, others, like WS and RH, may not be as impactful under the tested conditions.

Moreover, based on R-squared values, the results indicate that the inclusion of SI in the forecasting models is significantly associated with increased model performance ($p < 0.001$), as evident from the much higher median R-squared value (93.50%) compared to those without it, which have a lower median R-squared value (86.84%). This signifies the substantial impact of solar irradiance as an input feature in SPVPO forecasting.

Also, models that include AT show superior performance, with a high median R-squared (95.34%), compared to models without AT features, which exhibit a lower median R-squared (91.97%). The result of the Mann–Whitney test further confirms the significant difference in the R-squared values between the two groups of models ($p = 0.002$). Similarly, including WS may also lead to a substantial improvement in model accuracy ($p = 0.045$), with models featuring WS achieving a higher median R-squared (91.09%) compared to those without WS, which have a lower median R-squared (89.91%).

Including PPO ($p = 0.039$) and RH ($p = 0.022$) also results in statistically significant improvement in R-squared values. Models with PPO show a higher median R-squared of 91.63% than those without PPO with lower (87.42%); also, models with RH have a higher median R-squared of 90.36% compared to 88.27% for models excluding RH.

The statistical analyses using the Mann–Whitney U test on SPVPO forecasting models demonstrate that including specific features is associated with significant differences in both MAPE and R-squared values, indicating potential enhancement of model performance. SI, AT, and PPO are shown to be crucial in enhancing model performance, with pronounced differences in both metrics when included. Including at least one of these three features significantly lowers MAPE and increases R-squared values. On the

contrary, WS and RH show a less dramatic yet statistically significant improvement in R-squared values, unlike in MAPE. These results clearly emphasize the beneficial impact of integrating specific environmental and historical data features into SPVPO forecasting models, with SI, AT, and PPO standing out as particularly influential.

Furthermore, Table 4 shows the statistical comparison results of WTPO forecasting models with and without the identified important features—WS, PPO, AT, WD, and TS—in terms of their MAPE and R-squared values. It illustrates the potential impact of various features on WTPO forecasting accuracy, as measured by MAPE, along with supplementary results from the Mann–Whitney U test. The test results are presented in a structure similar to the previous table, facilitating comparisons between SPVPO forecasting models. Contrary to expectations, the median MAPE values for models with WS and/or WD are at par with those without WS and/or WD. The statistical tests also do not provide statistical evidence of a significant difference ($p > 0.1000$). This is also observable when wind PPO is included in the model ($p = 0.317$).

Examining the potential impact of the inclusion of TS in the model, the median MAPE for models with TS (13.15%) is far lower than those without TS (33.15%). The statistical test, however, does not provide statistical evidence for this ($p = 0.122$). One probable factor here is the limited number of models with TS, which may lead to unreliable interpretations of the large discrepancy in the median values and the borderline significance presented by the test itself. Notably, the median MAPE value of models with AT (8.11%) is lower than that of models without such features (13.00%). The statistical test also confirms the presence of significant differences in forecasting accuracy ($p = 0.019$), indicating a strong potential to improve the model's performance significantly.

In terms of their R-squared values, models including WS significantly outperform those without it ($p = 0.005$), as indicated by a higher median (96.70%) compared to the latter, which has a relatively lower median (91.30%). This highlights the critical role of WS in predictive accuracy in WTPO forecasting.

Significant discrepancies in the median R-squared values can also be seen when PPO and/or TS are included in the model. Models with PO have a higher median R-squared (97.31%) than those without PPO, which have a lower R-squared (91.30%). The Mann–Whitney test provides statistical evidence of a significant difference between the R-square values ($p < 0.001$). Similarly, models with TS have a significantly ($p < 0.001$) higher median R-squared (96.50%) than those without TS which have a lower R-squared (87.71%). This highlights the importance of PPO and turbine operational characteristics in enhancing forecasting accuracy.

On the other hand, including AT and/or WD may not significantly improve the model, as evidenced by the relatively small discrepancies between the median R-squared values. Specifically, the median R-squared of models with AT is 95.00%, which is not significantly ($p = 0.738$) superior to 96.96% for those without AT. The median R-squared of models with WD is 90.64%, which seems higher than 87.65% for those without WD, but this difference is not significant ($p = 0.413$).

The statistical data analyses from two tables on WTPO forecasting show that the inclusion of WS, PPO, AT, and TS is particularly beneficial in enhancing both the accuracy and explanatory power of models. WS and TS, notably, lead to substantial improvements in R-squared values, indicating a more robust model fit. PPO

consistently reduces forecasting errors and increases the variance explained, emphasizing its predictive value. Additionally, AT significantly reduces forecast errors but does not substantially improve the explanatory power of models, suggesting its role in specific aspects of forecasting accuracy.

These findings imply that WTPO forecasting models perform better when WS, PPO, TS, and/or AT are included. However, the results above do not demonstrate the significance of WD in forecasting WTPO. In particular, the inclusion of WD was not observed to be directly associated with improved model performance. This may be due to the underlying interconnection between WS and WD, which affects WTPO. To check for the probable direct and indirect effect of WD on WTPO, the joint inclusion of WS and WD into the model was investigated. Table 5 shows the results of the statistical comparison of WTPO forecasting models based on joint inclusion of WS and WD—with both WS and WD, with WS only, and without both—using the Kruskal–Wallis H test using MAPE and R-squared.

The statistical analyses of MAPE and R-squared values in WTPO forecasting models show significant variability based on the inclusion of WS and WD. For MAPE, models lacking both WS and WD exhibit higher median MAPE values (15.62%) than those with both WS and WD and those with WS only. Kruskal–Wallis test provides statistical evidence of significant differences among the groups ($p=0.035$). Through the Mann–Whitney test, post hoc analysis reveals that the source of this difference is the significant difference between models with both WS and those without both ($p=0.019$). Although the direct and indirect impact of wind direction on enhancing model performance is not explained, this still provides statistical evidence of a significant difference in model performance between models with and without these features.

On the other hand, R-squared values substantiate the impact of these features on forecasting performance. Models without WS and WD demonstrate relatively lower median R-squared values (91.30%). Meanwhile, models with WS and WD as inputs have a higher median (95.32%) than those with WS alone, with the highest median metrics (97.60%). The median R-squared values do not have an intuitive pattern, especially between models with WS and WD (95.32%) and models with WS alone (97.60%). Including more input variables in the model is expected to improve the R-squared value. However, this discrepancy may be attributed to the diversity of ML methods and the quality of the data available. Nonetheless, the Kruskal–Wallis test shows that there are significant differences in the median R-squares across different

Table 5 Statistical comparison of MAPE and R-squared according to joint inclusion of wind speed and/or direction in forecasting WTPO using Kruskal–Wallis test

Model	MAPE			R-Squared		
	Count	Median	<i>p</i> value	Count	Median	<i>p</i> value
Without WS & WD	49	0.1562 ^b	0.035	26	0.9130 ^a	0.003
With WS only	17	0.1350 ^{a,b}		91	0.9760 ^b	
With WS & WD	142	0.0952 ^a		84	0.9532 ^c	

^{a,b,c} Different letters indicate significant differences in model accuracy between the groups based on Mann–Whitney test

models ($p = 0.003$), and pairwise comparisons further reveal that the R-squares across all three models are indeed significantly different from each other ($p < 0.025$).

These findings underscore the importance of a careful assessment of feature selection in developing more precise and robust forecasting models for WTPO.

Machine learning-based forecasting techniques

The ML techniques and models widely used for SPVPO and WTPO forecasting were itemized and ranked according to the frequency of use in the reviewed literature. Figure 5 provides an overview of the distribution of ML-based techniques and models commonly used in SPVPO and WTPO forecasting. The overall distribution of machine learning models used in SPVPO and WTPO forecasting studies showcases a variety of approaches with diverse potential to achieve optimal model performance. Among these, hybrid models and neural networks, which constitute approximately 70% of the total models evaluated, are the most commonly used machine learning models. Specifically, hybrid models make up approximately 37.7% of the total models, whereas neural networks encompass 28.7% of the total applications. Ensemble methods and regressors are less frequently employed but still notable, comprising over 10% of the models. SVMs are the least utilized, representing 8.2%. This breakdown indicates a strong preference for hybrid models and ANNs in forecasting SPVPO and WTPO due to their ability to leverage multiple learning algorithms to improve performance. It also highlights the diverse range of other models employed to handle the complex problem of energy forecasting.

Furthermore, Fig. 6 shows the distribution of retrieved literature across various machine learning-based forecasting models identified for SPVPO and WTPO. As expected, ANN is the primary ML model commonly used for forecasting SPVPO and WTPO, accounting for over 60% of the literature. As previously discussed, all other machine learning models identified were also deemed to be commonly used in the domain. WTPO forecasting studies have explored the use of ANN and its hybridization with traditional and enhanced optimization algorithms, as well as other ML techniques.



Fig. 5 Distribution of ML models for SPVPO and WTPO forecasting literature

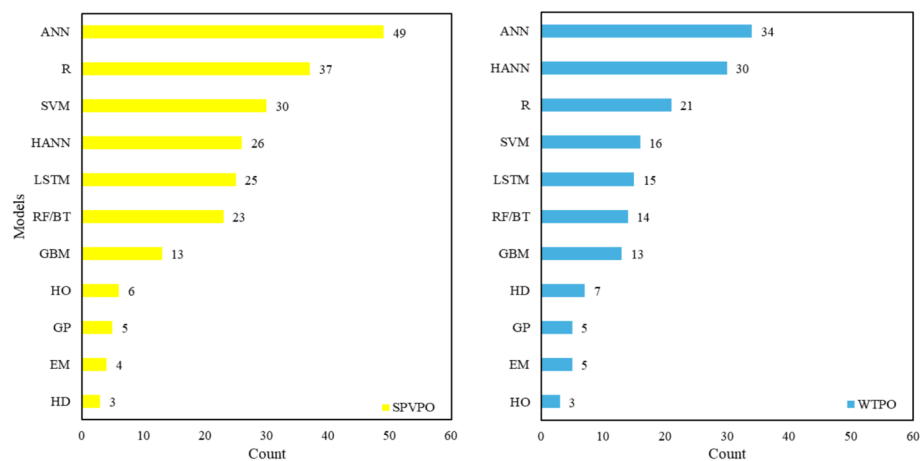


Fig. 6 Summary of ML-based models employed in SPVPO (A) and WTPO (B) forecasting studies

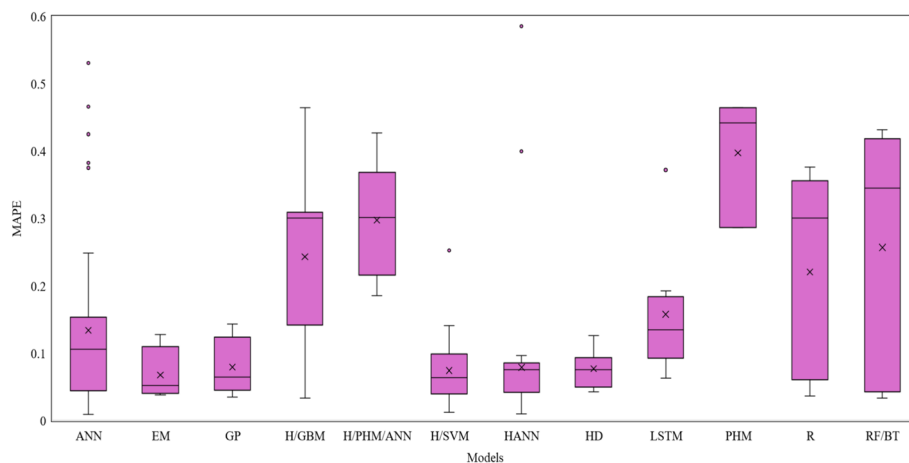


Fig. 7 Forecasting Accuracy of Various ML Models and Techniques Used for SPVPO and WTPO Forecasting in terms of MAPE

Another ML technique that has captured the current interest of SPVPO and WTPO forecasting studies is decomposition-based approaches such as empirical mode decomposition, wavelet decomposition, and Fourier transform. These ML techniques and models were then evaluated in terms of their forecasting accuracy, as measured by MAPE and R-squared (Figs. 7 and 8). The previous subsection focused on the overall performance of the models, with or without the top input features used in the domain, regardless of the actual ML model and techniques employed.

Figure 7 presents multiple boxplots that illustrate the distribution of MAPE values across various ML models, offering insights into their central values and variability in forecasting accuracy. Among them, ensemble models, GP models, and SVMs and hybrid SVMs demonstrate the lowest median MAPE values—5.19%, 6.40%, and 6.32%, respectively—followed by hybrid ANNs and decomposition-based hybrid models. Notably, the small width of their respective box plots suggests that these models perform more consistently across different model applications. The narrow spread in MAPE values

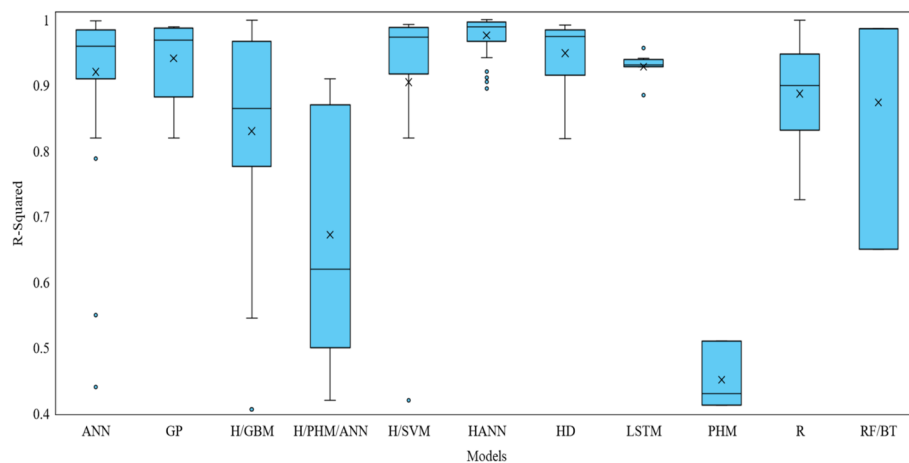


Fig. 8 Forecasting accuracy of various ML models and techniques used for SPVPO and WTPO forecasting in terms of R-squared

indicates the reliability and consistency of such models in obtaining accurate forecasts, which are preferable in applications.

Additionally, ANNs generally have a lower median MAPE (10.60%). ANNs also have the potential to outperform the models mentioned above in specific applications. LSTM is another ML model with a relatively smaller median MAPE value (13.50%) and variability, indicating its effectiveness in handling complex datasets with reasonably consistent results.

All other models have far higher median MAPE values with high variability. Models such as PHM (44.14%) and RF and Bagged Trees (34.50%) exhibit high median MAPE values, suggesting weak overall performance. However, RF and bagged trees also show the most considerable variability in their MAPE values, indicating inconsistency in their forecasting accuracy. This means that while the model can perform exceptionally well under certain conditions, its performance can fluctuate significantly. ANN-PHM hybrid models also do not have a desirable overall performance based on the retrieved studies.

Moreover, Fig. 8 displays the distribution of R-squared values across various ML models, utilizing multiple boxplots to highlight differences in performance stability and central tendency among the models employed in forecasting. Hybrid ANNs (98.88%) and GP models (96.88%) show exceptionally high median R-squared values with very low variability. This indicates not only superior performance but also consistency across different implementations. Similarly, decomposition-based hybrid models achieve a high median R-squared (97.46%) with moderate variability, suggesting robustness in complex forecasting scenarios. ANNs, ensemble models, and (hybrid) SVMs maintain high median performances with relatively low variability, showcasing their efficacy and reliability in stable forecasting.

LSTM stands out for its extremely low variability, paired with a high median (93.14%), indicating consistent outputs ideal for tasks requiring reliable, time-sensitive predictions. However, this result may be influenced by a few factors in this comparison. These results highlight the importance of selecting appropriate ML models based on the forecasting task's nature and the desired consistency in outcomes. Conversely, models such

as PHM and its hybrid with ANNs display significantly lower median values; ANN-PHM hybrid models show the highest variability among all models. This indicates poorer overall performance and high variability, which might make them less reliable for consistent application. RF and bagged trees also exhibit a wide range of performances despite a decent median (81.80%), reflecting the potential for high returns under specific conditions but generally unpredictable results.

In terms of MAPE and R-squared, ensemble models were superior among the ML models, followed by GPs, SVMs, hybrid ANNs, and decomposition-based hybrid approaches. The results demonstrate that ensemble and hybrid models have enhanced forecasting accuracy, aligning with the findings of related studies, such as [62] and [63]. SVM, GP, and ANN were also observed to have desirable forecasting accuracy due to their capability to capture the complex and nonlinear behavior of solar PV and wind energy. Decomposition-based approaches such as empirical mode decomposition, wavelet decomposition, and Fourier transform were also shown to enhance the forecasting accuracy of the models [64]. Decomposing the data into multiple sections before training the ML model can lead to improved model performance.

Machine learning models and techniques can achieve optimal forecasting accuracy by maximizing their capabilities in, but not limited to, learning complex and nonlinear patterns, effectiveness in high-dimensional data, robustness against overfitting, use of feature importance, capturing long-term dependencies, and optimal tuning of hyperparameters, if applicable. However, apart from optimal model performance, selecting the appropriate and efficient ML models should consider other aspects such as complexity, interpretability, and computational efficiency. Table 6 summarizes the advantages and disadvantages of these machine learning techniques in energy forecasting based on various model configurations and considerations. This highlights the abilities and limitations of the identified ML models, which can help choose the optimal ML model depending on the nature of the data and study.

Selecting the suitable machine learning model for renewable energy forecasting requires careful consideration of its implications, trade-offs, and practicality. ANNs offer a versatile framework for forecasting complex patterns; however, their deep structures can impact computational efficiency and interpretability. LSTM networks capture

Table 6 Advantages and disadvantages of ML models in forecasting SPVPO and WTPO

Characteristics	ANN	SVM	RF	GBM	LSTM	GP	HM	EM
Complex pattern learning	✓					✓	✓	✓
High-dimensional effectiveness		✓	✓					✓
Robustness to overfitting		✓	✓				✓	✓
Feature importance			✓	✓		✓		✓
Forecasting accuracy			✓	✓				✓
Interpretability				✓		✓		
Captures long-term dependencies					✓			
Memory intensive		✓	✓			✓		✓
Computational complexity	✓			✓	✓	✓	✓	✓
Training runtime	✓		✓		✓			✓
Hyperparameter tuning difficulty		✓		✓	✓	✓		✓

long-term dependencies in sequential data but require careful parameter tuning and significant computational resources. SVMs provide robustness against overfitting and handle high-dimensional data efficiently; however, the manual selection of optimal kernel functions can pose a computational complexity issue. Ensemble methods combine multiple models to improve forecasting accuracy, but they can be resource-intensive and require longer training time. By carefully evaluating the technical aspects of various models, researchers can make informed decisions to optimize model performance and achieve desired outcomes in renewable energy forecasting.

Challenges and opportunities

Despite their promising potential in the field, machine learning-based forecasting models face several challenges. These may include data scarcity, data quality, model complexity and interpretability, and integration into energy management systems. This study identified the ongoing challenges in forecasting SPVPO and WTPO.

Data availability

In forecasting SPVPO and WTPO, various data sources are utilized, including in situ, reanalysis, satellite-based or remotely sensed, physics-based, and hybrid data. Recent studies highlight the importance of in situ data as a highly preferable data source for forecasting solar and wind energy, emphasizing its accuracy, reliability, and suitability for capturing local variations in weather conditions. However, data sources for forecasting SPVPO and WTPO, such as weather stations, satellite imagery, and sensor networks, may be sparse and irregularly distributed across geographic regions. Limited coverage of monitoring infrastructure can lead to gaps in data availability, hindering the development of accurate forecasting models, particularly in remote or less developed areas. Benitez et al. [31] and Ibañez et al. [32] analyzed the accuracy of forecasting models for SPVPO in the Philippines with solar irradiance as an input variable collected from various data sources. Both studies demonstrate that forecasting using in situ data continues to yield optimal model performance. However, remotely sensed and reanalysis of solar irradiance data can also yield competent forecasting accuracy when in situ data are unavailable. Specifically, [32] demonstrated that forecasting SPVPO using SARIMAX with in situ solar irradiance data still yields the highest accuracy among various data sources, especially during rainy and hot dry seasons; no sufficiently significant difference in forecast accuracy was observed during the cold, dry season. Alternatively, without in situ data, ERA5 reanalysis data are preferable for dry seasons, while Himawari-8 satellite data are preferable for rainy seasons. Similarly, [31] trained SARIMAX models to forecast SPVPO across different data sources. This shows that the model performs best when in situ solar irradiance is used as input, followed by ERA5 reanalysis data and Fengyun-4A satellite data.

Based on the literature, the majority (33.61%) of the ML-based forecasting models for SPVPO and WTPO were trained using in situ data, followed by NWP-based (25.42%) and remotely sensed data (17.65%). Few models were trained using a hybrid of in situ and NWP-based data (7.98%), reanalysis data (7.56%), and simulated data. A statistical comparison of the models using these data sources was carried out to support this. Figure 9 compares the model accuracy, as measured by MAPE and R-squared, in forecasting

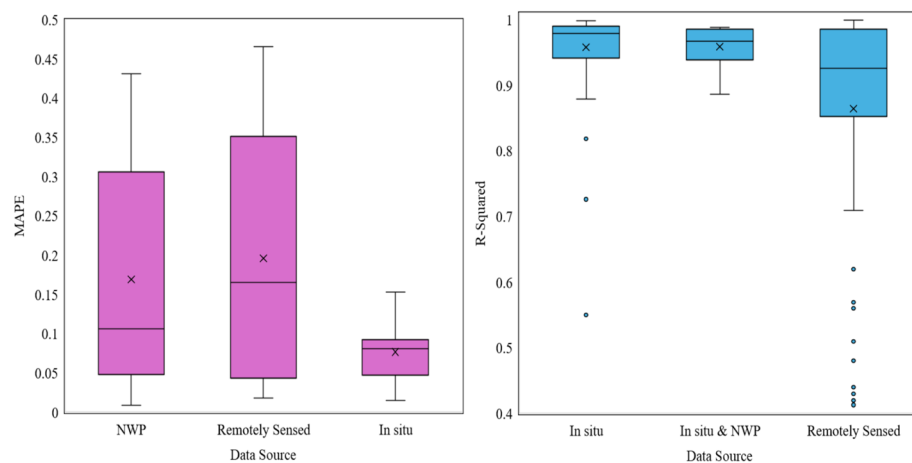


Fig. 9 Forecasting accuracy of ML models across various data sources for SPVPO and WTPO forecasting in terms of MAPE and R-squared

SPVPO and WTPO across various data sources. Data sources included in the analysis were identified in the reviewed studies: in situ, NWP-based, remotely sensed, and in situ-NWP hybrid data. Some data sources, such as reanalysis data, were excluded due to the small group size, which may have led to unreliable conclusions.

Based on the results, forecasting using in situ data generally yields optimal model performance, with median MAPE (8.11%) and R-squared (97.89%), with the narrow spread in MAPE and R-squared values, such data yields forecasting reliability and consistency with high accuracy. The combined use of in situ and NWP-based data also yields high model accuracy regarding R-squared (96.75%), which is also paired with low variability. Moreover, using NWP-based data generally yield high forecasting accuracy in terms of MAPE (10.60%), followed by remotely sensed data (16.51%). With the controlled environment and biased assumptions inherent in NWP-based computations and simulations, the complexity, variability, and nuances typically observed in actual or in situ data may not be accurately represented. Hence, accurate forecasting is highly expected. The Kruskal–Wallis test and post hoc analysis results do not provide statistical evidence that the median MAPE values of models trained using NWP-based and remotely sensed data differ significantly; instead, they emphasize the superiority of using in situ data. In terms of R-squared, the results of the statistical tests also confirm that training the models using in situ data is superior compared to those using other data sources, particularly remotely sensed data. These results underscore the importance of using in situ data in forecasting SPVPO and WTPO, which aligns with the results of existing studies. Other data sources, such as remote sensing, can be used as alternatives for training forecasting models; however, their accuracy may not be comparable to that achieved when in situ data are used.

Data quality

Forecasting SPVPO and WTPO relies heavily on meteorological data, including solar irradiance, wind speed, temperature, and atmospheric pressure. Inaccuracies in these meteorological measurements due to factors, such as sensor inconsistency, calibration

errors, or data processing issues, can lead to erroneous forecasts and poor model performance. Solving such data quality issues is one of the crucial challenges that can introduce biases and uncertainties into the forecasting models, adversely affecting their reliability and accuracy.

In the reviewed studies, one potential source of data quality issues is the use of alternative data sources (e.g., remote sensing and numerical weather prediction models). The significant difference in forecasting accuracy between the models trained using alternative data sources and in situ data may be attributed to their discrepancy with the actual data. The literature provides sufficient evidence that using in situ data is superior among the data sources. Before comparing model accuracy, [31] compared the reanalysis and satellite-derived solar irradiance data to in situ solar irradiance data using simple linear regression analysis. The results showed a weak linear association between the remotely sensed data and the in situ data. However, Fengyun-4A solar irradiance data, having the highest R-squared value, provides the smallest overall differences from the in situ solar irradiance data. This may be the primary reason for its superiority among alternative data sources in terms of model accuracy (i.e., RMSE).

Another data quality concern in the domain is the presence of missing values. Associated with data availability issues, missing data may arise due to, but not limited to, sensor and communication errors or environmental factors affecting the solar panels or wind turbines. In the reviewed studies, missing values are commonly excluded from the data or imputed. With this, missing values can distort the underlying patterns and relationships in the data. Since renewable energy exhibits strong temporal patterns driven by weather conditions, meteorological fluctuations, and time variables such as time of day and seasonality, this concern complicates the forecasting process in capturing fluctuations and variability in power production over time. Also, selecting an appropriate imputation technique is challenging due to the unique characteristics of such renewable sources. Applying an inappropriate imputation method can induce bias and uncertainty in the data.

Model complexity and interpretability

As highlighted in the previous section, a trade-off exists between model complexity and interpretability in forecasting SPVPO and WTPO. This is crucial in deploying and integrating the forecasting models into the energy management system. Simple models, such as linear regression models and decision trees, provide greater interpretability but may not accurately capture the complexity of the data. Meanwhile, complex models, like deep learning techniques, may lack transparency and interpretability despite achieving higher forecasting accuracy. Balancing this trade-off requires careful consideration of the specific requirements and constraints of the forecasting process, as well as the preferences and priorities of the stakeholders involved in decision-making.

Uncertainty and variability of weather and energy sources

Another challenge in the domain is accurately capturing uncertainty and variability in the data. SPVPO and WTPO production are consistently associated with uncertainty and variability attributable to, but not limited to, weather fluctuations, cloud cover, and

wind speed variations. Forecasting models must account for this uncertainty to provide reliable predictions, especially for short-term and intraday forecasting horizons.

Integration into energy management systems

Operational integration and decision support are crucial components of integrating forecasting for SPVPO and WTPO into energy management systems. These involve seamlessly integrating forecasting outputs into EMS workflows and providing stakeholders with timely and actionable insights derived from forecasting models to support real-time decision-making and planning, as well as optimizing energy generation, storage, and distribution strategies. However, operational integration and decision support may be challenging since solar and wind energy generation is inherently variable and can be influenced by weather conditions, environmental changes, and system dynamics.

Forecasting models must provide reliable predictions at different time horizons, ranging from short-term forecasts for intraday energy scheduling to long-term forecasts for capacity planning and resource allocation. Ensuring the accuracy and reliability of forecasting outputs in real-time operational settings is crucial for EMS users to make informed decisions and effectively manage energy resources.

Additionally, EMS users require user-friendly interfaces and intuitive visualization tools for easier accessibility and interpretability of forecasting results. Integrating forecasting output with EMS systems, such as energy market platforms, demand response systems, and asset management tools, poses technical challenges related to data interoperability, system compatibility, and information exchange protocols. Developing seamless integration mechanisms and automated decision-support systems that leverage forecasting insights to optimize energy operations in real-time is essential.

Opportunities

Forecasting SPVPO and WTPO offers significant opportunities for revolutionizing energy management and advancing the integration of renewable energy sources into the power grid. While the challenges discussed above exist, addressing them presents several promising opportunities in the domain:

- *Big data and the internet of things.* With the domain's status, the proliferation of big data analytics and Internet of Things (IoT) technologies enables the collection of vast amounts of data from renewable energy assets, weather sensors, and grid infrastructure. This can resolve the data availability and quality concerns in the domain. Leveraging these data sources through machine learning can improve the accuracy and timeliness of forecasting models for SPVPO and WTPO.
- *Advanced data processing techniques.* Addressing data quality issues in forecasting SPVPO and WTPO aims to improve the data's integrity, reliability, and usability. One approach is to perform data cleaning and preprocessing techniques. Data cleaning techniques involve detecting and correcting errors, inconsistencies, and outliers in the dataset. This may include removing duplicate entries, correcting formatting errors, and imputing missing values using interpolation or averaging techniques. Preprocessing steps, such as normalization and scaling,

can also be applied to ensure that data are standardized and comparable across different variables and time periods.

In the reviewed studies, advancements were made in effectively performing and streamlining data cleaning procedures by incorporating specialized algorithms into the ML model. Fahim et al. [65] introduced a hybrid ML model for forecasting WTPO that automatically performs data preprocessing techniques: Histogram-Based Outlier Score (HBOS) for outlier detection and filtering and Genetic Algorithm (GA) for feature selection. Integrating such techniques into the ML model was shown to improve model accuracy. Here, the HBOS technique was also shown to be superior to Isolation Forest (IF), which was commonly used in earlier WTPO forecasting studies. It also provided novel insights into the positive impact of incorporating a wrapper-based feature selection algorithm, such as GA, into the model. Similarly, [66] proposed a hybrid power curve modeling technique that detects and removes outliers using fuzzy c-means (FCM) clustering, resulting in enhanced model performance.

Furthermore, advancements in data processing techniques have been made to resolve missing values. Benitez et al. [67] introduced the novel expansion of the column mean imputation method to address the issue of missing values in forecasting SPVPO. This data preprocessing technique was employed to fill the missing values in shortwave radiation from Himawari 8/9 satellite data and surface solar radiation downward from ERA5 reanalysis. This method was shown to improve forecasting accuracy, offering a better approach than simply excluding such missing values.

- Probabilistic forecasting and advanced ML techniques

The literature for forecasting SPVPO and WTPO is highly concentrated on point forecasting, which provides a single forecast at a particular time in future [68]. Given the uncertainty and randomness inherent in renewable energy sources, point forecasts may struggle to predict future values, resulting in poor model performance accurately. With this, probabilistic or interval forecasting is gaining momentum in forecasting renewable energy production; however, there is still a lack of literature on this [68, 69]. Future researchers can further explore the realm of probabilistic forecasting to capture the uncertainties and complex variability of solar and wind energy. Additionally, continuous advancements in machine learning techniques, such as deep learning, ensemble methods, and hybrid models, offer opportunities to develop more accurate and robust forecasting models for SPVPO and WTPO. These techniques can better capture complex relationships and nonlinearities in renewable energy generation processes.

In response to these challenges, incorporating domain knowledge into the modeling process, enhancing data quality and availability, and exploring innovative algorithmic approaches can be crucial. Hybrid forecasting approaches combining physics-based models with data-driven machine learning techniques offer opportunities to leverage domain knowledge and empirical data for improved forecasting accuracy. Integrating physical models with machine learning algorithms can enhance the interpretability and reliability of forecasting models.

Proposed framework for forecasting SPVPO and WTPO

This study highlighted the need for a standard framework for forecasting SPVPO and WTPO. After evaluating and analyzing the relevant studies in this domain, such a framework was designed to guide future researchers in exploring ML-based forecasting models for SPVPO and WTPO. Figure 10 shows an overview of the proposed forecasting framework.

The proposed forecasting framework for SPVPO and WTPO involves a general methodology that includes forecasting configuration, data collection, data preprocessing, feature selection, decomposition techniques, and data splitting. The forecasting objectives are identified and set based on the time horizon. Data are collected from reliable sources, preferably sensor networks, weather stations, and/or satellite data. SPVPO forecasting incorporates solar irradiance, ambient temperature, wind speed, and relative humidity, while WTPO forecasting includes wind speed, wind direction, ambient temperature, atmospheric pressure, and relative humidity.

Data are then preprocessed using traditional or advanced methods, including outlier detection and filtering, data imputation to fill missing values, normalization or scaling of variables for faster convergence, feature selection or reduction techniques such as correlation and PCA, and decomposition techniques like EMD, enhanced EMD, wavelet decomposition, and Fourier transformation. The processed data are then sequentially split into train, validation, and test data. For large datasets, k-fold or time-series cross-validation is performed, while simple sequential splitting is done for small datasets.

ML models can be used for point and probabilistic forecasting, with various models being utilized for different data types. ANNs, SVMs, regressors, and hybrid and ensemble models are recommended for point forecasting, while GPs, quantile regression, and ensemble models are recommended for probabilistic forecasting. Conventional time series analysis or advanced machine learning techniques, such as LSTM networks and CNNs, can be used for sequential data. High-dimensional data, including prior power output and meteorological variables, can be processed using ANNs, SVMs, RFs, and hybrid and/or ensemble models.

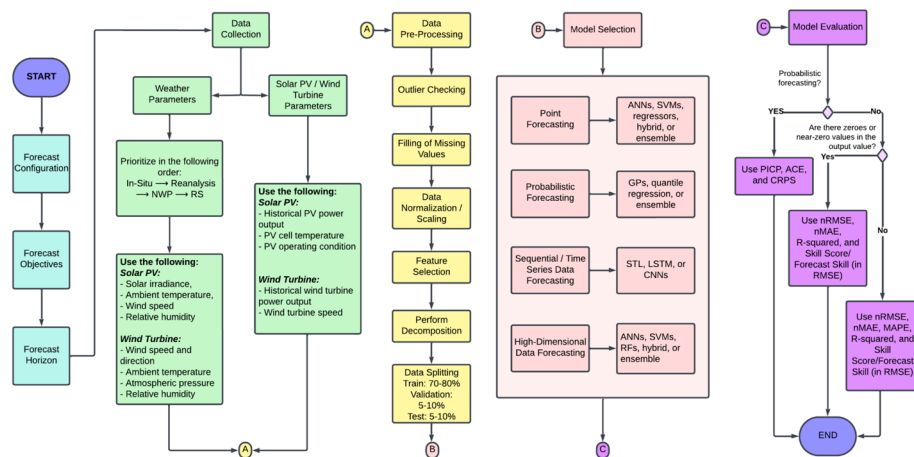


Fig. 10 Overview of the proposed forecasting framework

Lastly, it is recommended to perform model evaluation using the commonly used metrics, such as RMSE, MAE, MAPE, R-squared, and Skill Score/Forecast Skill for point forecasting, and PICP, ACE, and CRPS for probabilistic forecasting. Scale-dependent and scale-free metrics should be balanced, and the model should be evaluated using metrics that are appropriate for each domain.

Case study

This section evaluated current publications on SPVPO and WTPO forecasting in the Philippines. The analysis provided a general overview of the literature and its alignment with the proposed framework, identifying areas for improvement and suggesting future studies in the Philippines.

The literature on SPVPO and WTPO in the Philippines is limited. Hence, for this study, a more relaxed literature selection was conducted, encompassing relevant studies within the broader domain of forecasting the country's solar and wind power output. The Google Scholar Database was also utilized. After reviewing the literature, eleven (11) pieces of literature related to forecasting SPVPO were retrieved; all were associated with short-term, direct, and point forecasting. Meanwhile, only four (4) pieces of literature are related to point forecasting WTPO, which applies various forecasting horizons; two employed indirect forecasting through wind speed, while others used direct forecasting. These literatures were evaluated and analyzed using the same methodology applied in this study. Table 7 shows the results of the step-by-step evaluation of this literature in the Philippines, including the model accuracy metrics.

Consistent with the results of this study, RMSE (81.8%), MAE (54.5%), and MAPE (45.5%) were the accuracy metrics commonly used in forecasting SPVPO in the Philippines. Due to the lack of sufficient literature, statistical tests are not viable to determine significant differences in the accuracy of the forecasting models. Hence, this case study performed a simple comparison of the forecasting models and techniques.

Regarding the inputs considered in the forecasting process, the top variables for forecasting SPVPO in the Philippines were identified—SI, AT, WS, and RH—which align with the results of this study. 72.7% of the reviewed literature considered these input variables. Additionally, TP (90.9%), PPO (63.6%), and WD (60.0%) were also considered by most of these literatures; few of these literatures considered CC (30.0%). Notably, one literature review considered PV operating characteristics, including back-cell temperature (BCT) and tilt angle (TA).

In terms of the models utilized, half of the literature related to forecasting SPVPO employed ML models (in particular, ANNs, LSTM, and XGBoost); 40.0% of the literature incorporated statistical models (i.e., SARIMA, SARIMAX, and ANCOVA); and one study compared statistical, ML, and ML-statistical ensemble models. Specifically, SARIMAX is the model widely used for forecasting SPVPO in the Philippines, followed by ANNs and LSTM. The seasonal autoregressive integrated moving average with exogenous variables (SARIMAX) model is a statistical time-series forecasting technique that captures autoregressive, moving average, and integration components, as well as seasonality and external factors.

The significance of weather variables in predicting SPVPO was emphasized by [70] and observed an evident decrease in MAPE values, except in Negros Occidental with

Table 7 Evaluation of SPVPO and WTPO forecasting literature in the Philippines

Source	Year	Horizon	Type	ML models	Input variables	Metrics used
<i>SPVPO forecasting</i>						
[70]	2024	Short term	Direct	XGBoost	PPO, SI, AT, WS, WD, RH, TP, CC	MAPE, SS
[71]	2024	Short term	Direct	LSTM	PPO, SI	RMSE, MAE, MAPE
[72]	2024	Short term	Direct	PCA-ANCOVA	SI, WS, WD, RH, AT, TP, BCT, TA	Cor
[32]	2023	Short term	Direct	SARIMA, SARIMAX	PPO, SI, WS, WD, RH, AT, TP	RMSE, MAE, MAPE
[67]	2023	–	Direct	XGBoost	PPO, SI, WS, WD, AT, RH, TP	RMSE, MAE
[73]	2023	Short term	Direct	SARIMAX, LSTM, XGBoost, Ensemble	PPO, SI, AT, WS, WD, RH, TP, CC	RMSE, MAE, MAPE, nRMSE, nMAE
[74]	2023	Short term	Direct	FFBPNN	SI, AT, WS, RH, TP	MSE
[75]	2023	Short term	Direct	RFR, PCR	TP, AT, WS, WD, RH, CC	RMSE, R-squared
[76]	2022	Short term	Direct	SARIMAX	PPO, SI, AT, WS, WD, RH, TP, CC	RMSE, MAE
[76]	2022	Short term	Direct	SARIMAX	PPO, SI, AT, WS, WD, RH, TP	RMSE, MAE
[77]	2020	Short term	Direct	ANN	SI, AT, WS, RH, TP	MSE, MAPE
<i>WTPO forecasting</i>						
[78]	2023	Short term	Indirect	SVR, EBT	T, WS, AT, RH, AL	RMSE, MAE, MAPE
[79]	2023	Long term	Indirect	SMA, ARIMA, SARIMAX, LR, RR, LASSO, KNNR, DT, GBR, XGBoost, RF	T, AT, RH, AL, WS	RMSE, MAE, R-squared
[80]	2019	–	Direct	Structured ANN	PPO	Cor
[81]	2019	Ultra-short term	Direct	PHM	WS	MSE

a small discrepancy. Benitez et al. [72] emphasized that weather variables like SI, AT, RH, and WS can be highly interconnected and correlated. Both studies employed PCA to simplify and resolve the complexity, dimensionality, and multicollinearity of the weather data. Using sensitivity analysis, [77] highlighted the significant importance of SI and AT in forecasting SPVPO, followed by TP, RH, and WS.

Additionally, [72] demonstrated that PV operating configurations and characteristics, such as TA and BCT, are significant predictors of SPVPO. It was shown that an optimal tilt angle for a PV panel (e.g., 10° in Muntinlupa City) can improve SPVPO production. Benitez et al. [71] employed a novel data processing practice using cloud optical thickness (CLOT)-adjusted satellite-based SI data from AHI-8/9. Benitez et al. [71] Furthermore, the performance of adjusting SI data from AHI-8/9 (SWR) and FY-4A (SSI) was examined; the results confirmed an improvement in model performance due to CLOT adjustment of SWR but not that of SSI. Calibration or further adjustment of remotely sensed data, such as SI, by capturing key patterns and characteristics related to meteorological factors to achieve more precise predictions.

Moreover, the simulation in [67] demonstrated that data imputation significantly enhances model accuracy, even with 20% missing values, as evidenced by the increase in error metrics. The novel modified column mean imputation (CMI) substantially improved the model performance by enhancing the quality of each input data, SSRD,

and SWR. Benitez et al. [73] Also, the modified CMI was applied to fill data gaps, ensuring data reliability, completeness, and consistency for forecasting purposes.

The importance of careful evaluation of the input variables was highlighted by [77] and [82] before model training. The results show that the model that underwent feature reduction outperformed the one incorporating all inputs without filtering. Therefore, including more input variables in the model does not necessarily imply optimal model performance.

Also, [31] compared the accuracy of SARIMAX models for SPVPO across different data sources—in situ, reanalysis, and remotely sensed data. This showed reanalysis and FY-A4 satellite-based data were alternatives for unavailable in situ data. This can be due to their moderately strong correlations with in situ data, especially FY-A4 solar irradiance. Likewise, [32] suggested using SWR or SSRD (reanalysis) as a competitive alternative to in situ data. SSRD (in SARIMAX) is preferable for hot, dry and rainy seasons over SWR; SARIMA is sufficient for cool, dry seasons.

The current landscape of SPVPO forecasting in the Philippines may lack the application of advanced machine learning models and techniques. The relevant studies focus on SARIMAX models, which are statistical models based on linear regression and may struggle to capture complex nonlinear relationships between input and the target variables. In contrast, machine learning techniques, such as ANNs, can learn nonlinear patterns and potentially provide better accuracy for highly nonlinear data. Velasco et al. [74] and Velasco et al. [77] developed ANNs in forecasting SPVPO in the country; however, they may be limited to in-depth exploration of deep learning, hybrid, and/or ensemble models. Meanwhile, [73] compared statistical (SARIMAX), ML (LSTM, XGBoost), and their ensemble models; however, the best models identified do not include LSTM. Reference [71] also applied LSTM to examine the impact of solar irradiance data.

With these, ML-based forecasting SPVPO and WTPO literature in the Philippines must further delve into advanced and diverse models and techniques employed globally.

Application of the proposed framework to existing local literature

To evaluate the existing landscape of forecasting SPVPO and WTPO in the Philippines, the current practices employed were assessed based on the forecasting framework proposed in this study. Based on this assessment, areas for improvement and future studies in the locale can be identified and explored. Due to the limited literature on forecasting WTPO, this section mainly discusses the literature related to forecasting SPVPO. In terms of Forecasting Configuration, Data Collection, Data Preprocessing, Model Selection, and Model Evaluation, the alignment of the existing studies with the proposed framework is itemized:

(1) Forecasting configuration

All the local studies for forecasting SPVPO developed short-term point forecasting models. For a more effective EMS, the literature in this domain should offer diverse forecasting horizons, ranging from ultra-short-term to long-term forecasting, to substantiate ML-based forecasting outputs across different objectives in the local EMS.

(2) Data collection

Almost 80% of the SPVPO forecasting literature in the Philippines simultaneously considered and incorporated the input variables deemed significant in enhancing the accuracy of the forecasting models for SPVPO. These are solar irradiance, ambient temperature, wind speed, relative humidity, and prior PV output. Since most of the literature employs SARIMAX, time variables are unnecessary in this case.

In terms of data source, the majority (40.0%) of the literature used reanalysis data, followed by *in situ* (20.0%) and a combination of reanalysis and remotely sensed data (20.0%). Very few used remotely sensed data or a combination of the three. With the ongoing challenge of data availability for forecasting SPVPO, local studies may opt to collect data from competitive alternative sources such as reanalysis and satellite data. However, it has been shown that these data may not be as accurate as *in situ* data; hence, the forecasting accuracy of models trained using such data may be compromised. To address this, some local studies utilized calibrated data, such as CLOT-adjusted SWR data, presenting promising results. To address these issues, data availability must be compensated for, but not limited to, through calibration of reanalysis and/or remotely sensed data against other factors, such as cloud cover or geomorphological attributes.

(3) Data preprocessing

As a common practice, most literature has employed outlier detection and filtering methods to enhance data quality. Also, issues of missing values were resolved using data imputation techniques, such as the novel modified CMI method [67] or the use of gap imputing algorithm [83]. Normalization and scaling techniques, such as Min-Max normalization, were used as part of the data preprocessing task to achieve faster ML model and algorithm convergence. In practice, the normalization of historical SPVPO and/or the input variables can be incorporated inside the ML algorithm and, in some cases, reverted to the original scale, such as in [77].

Moreover, feature evaluation and selection were also incorporated in the literature. Pearson correlation is the standard feature evaluation and selection technique utilized, where variables highly correlated with SPVPO were highlighted and/or selected as final inputs in the forecasting model. Another technique applied in the literature is principal component analysis (PCA), which minimizes the dimensionality of the dataset (i.e., the number of input variables), thereby avoiding memory-intensive computations, model overfitting, and overparameterization. Gavina et al. [70] and Benitez et al. [72] transformed the input variables into principal components with smaller dimensionality. Benitez et al. [72] emphasized the need to solve multicollinearity issues arising from highly correlated input variables, evident among weather variables; PCA resolves this issue seamlessly. On a more straightforward approach, other literature, such as [32] and [73], employed feature selection via the VIF criterion, which detects the variables causing multicollinearity.

Since most studies used SARIMAX, time-series decomposition was performed, a common practice before building time-series models. For the same reason, most of

these studies split their datasets into train and test data with an 8:2 ratio. Meanwhile, a three-way data splitting, including validation data, was carried out in some literature incorporating complex models, such as [71] and [77]. In practice, validation data are used for hyperparameter tuning, which is ideal for developing complex models such as ANNs and XGBoost. In some cases, such as forecasting with cross-validation, validation data are unnecessary. Also, [70] applied a k -fold cross-validation approach for improved generalizability of the models.

In terms of data preprocessing, these studies align with the proposed framework. While the existing practices present promising approaches and techniques, other advanced and automated methods of outlier detection and filtering (e.g., HBOS, IF, FCM clustering), data imputation (e.g., EM algorithm), feature selection (e.g., GA), and decomposition techniques (e.g., EMD, enhanced EMD, wavelet decomposition, Fourier transform) can be further explored and applied to improve data quality before the development of forecasting models for SPVPO (and also WTPO) in the Philippines.

(4) Model selection

As discussed above, most studies on forecasting SPVPO in the Philippines have employed SARIMAX, followed by ANNs such as LSTM. This is because of the nature of the data, which is highly sequential or time-dependent. SARIMAX models, which utilize linear regression and statistical approaches, may struggle to capture complex, nonlinear relationships in the data. However, the proposed framework in this study suggests that advanced ML techniques, such as artificial neural networks (ANNs), could offer superior accuracy by effectively handling nonlinear data patterns. Some local studies have explored ANNs for SPVPO forecasting, although they may not have thoroughly investigated deep learning, hybrid, or ensemble models. Additionally, LSTM was employed; however, it did not perform among the best. Ultimately, the domain of forecasting SPVPO (and WTPO) in the Philippines should further expand the application of advanced ML-based models to maximize the potential of such an approach in the local setting.

(5) Model evaluation

So far, the local forecasting models for SPVPO are aligned with the framework regarding the model evaluation metrics. RMSE, MAE, and MAPE, the top metrics recommended by the framework, are the most used error metrics in local studies. However, local studies should not rely solely on these metrics and should also include other scale-free metrics, such as Skill Score and R-squared, for a more holistic and comprehensive evaluation of the ML models.

The recent literature on SPVPO forecasting in the Philippines reveals a prevalent focus on short-term, point forecasting models, primarily utilizing statistical and machine learning models (SARIMAX and ANNs, respectively). This paves the way for diversifying the forecasting horizon from ultra-short-term to long-term and incorporating point and probabilistic forecasting models. While data collection incorporates

significant input variables and various data sources, challenges persist in data availability and accuracy, especially with alternative sources such as reanalysis and satellite data. Data preprocessing techniques, including outlier detection, imputation, normalization, and feature selection, are well established. However, there is potential for further exploration of advanced techniques to enhance data quality. While model training and evaluation align with recommended frameworks, there remains a notable opportunity to expand the application of advanced ML-based models, such as deep learning, hybrid, or ensemble models, to maximize accuracy and effectiveness. Diversifying forecasting horizons, incorporating additional evaluation metrics, and exploring advanced data preprocessing and modeling techniques can further advance the forecasting of SPVPO and WTPO in the Philippines, ultimately contributing to more effective energy management systems.

Implications for practice and policy

This study on ML-based forecasting techniques for SPVPO and WTPO production is imperative in providing practical implications of the findings for energy producers, grid operators, policymakers, and other stakeholders in the renewable energy sector in the Philippines and globally. Machine learning-based forecasting can inform decision-making processes related to energy system planning and operation. Policies can promote the development of forecasting tools that integrate with energy management systems, enabling utilities to optimize energy generation, storage, and distribution more effectively. This can lead to cost savings, improved resource allocation, and reduced reliance on fossil fuels. This paves the way for exploring interdisciplinary collaborations with experts from various fields, such as meteorology, climatology, and energy systems.

By addressing these policy implications, stakeholders can harness the potential of machine learning-based forecasting to accelerate the transition to a more sustainable and resilient energy future. These policy measures can foster innovation, investment, and collaboration in the renewable energy sector, ultimately driving progress toward climate goals and energy security objectives.

Limitations and future research

While the current study provides valuable insights into machine learning-based forecasting for SPVPO output and WTPO, it is essential to determine its limitations. This may involve scope limitations, selection bias, data quality, and interpretation bias. The study may not encompass more recent studies or advancements in the rapidly evolving domain of renewable energy forecasting and machine learning. The selection of studies included in the study may introduce bias, potentially overlooking relevant research or perspectives.

In particular, the reviewed literature focused primarily on ultra-short- and short-term forecasting; therefore, the generalizability of the study results may be limited. Moreover, the geographical considerations of the reviewed studies were also not representative of the global scale. In the case of the Philippines, almost all relevant literature employs short-term forecasting horizons and is highly focused on the Luzon area. Future local and international research can delve into generalizing the domain by capturing short- to long-term forecasting studies across the Philippines and the globe.

Conclusions and recommendations

Accurate energy forecasting is crucial for efficient energy system management and stable energy infrastructure, addressing the impending intermittent nature of renewable sources such as solar PV and wind turbines. This study explored the application of machine learning techniques in forecasting SPVPO and WTPO. This study holds immense value in advancing renewable energy research and applications by highlighting state-of-the-art techniques, challenges, and opportunities, as well as significant innovations and future directions in ML-based forecasting for these critical renewable energy sources. It also formulates a comprehensive five-phase forecasting framework for this purpose.

The results of this study highlight the potential impact of specific meteorological and operational variables on improving the forecasting accuracy of the models, particularly in terms of MAPE and R-squared. In forecasting SPVPO, solar irradiance, ambient temperature, and prior PV power outputs are essential features, followed by wind speed and relative humidity. Meanwhile, in forecasting WTPO, wind speed, turbine speed, and prior WTPO were found to play a crucial role in achieving optimal forecasting accuracy, followed by ambient temperature and wind direction. The careful selection of input features to be considered in the forecasting process plays a pivotal role in achieving better model performance.

In terms of machine learning models, ensemble models, SVMs, GPs, hybrid ANNs, and decomposition-based hybrid models were identified as having desirable forecasting accuracy with high reliability and consistency, which may gain interest in the domain. This study also highlighted the challenges in this domain, including data availability and quality issues, the trade-off between complexity and interpretability, the uncertainty and variability of renewable energy sources, and the difficulty in integrating the forecasting process into energy management systems. These challenges, however, present opportunities to develop innovative, proactive techniques and approaches in the domain. The body of knowledge for ML techniques can be further expanded to achieve more accurate, robust, and efficient forecasting models and methods for SPVPO and WTPO.

Given its abundant solar and wind energy resources, the Philippines actively adopts renewable energy sources, such as solar PV and wind turbines, to mitigate climate change and diversify its energy portfolio. While the current landscape aligns with the proposed framework, the domain of forecasting SPVPO and WTPO in the country needs more relevant literature, scope, and generalizability. By addressing these areas for future local and international research, the field of renewable energy forecasting using machine learning can overcome current limitations and capitalize on emerging trends to drive innovation and sustainability in the energy sector.

To advance machine learning-based forecasting for SPVPO and WTPO, future research should focus on standardization, scope, forecasting processes (data collection, preprocessing, model building, and evaluation), and integration processes. Opportunities in forecasting SPVPO and WTPO in the Philippines and globally include evaluating forecasting accuracy and reliability across various data sources and time horizons, comparing model performance using different exogenous variables and data sources, developing probabilistic forecasting models, investigating the impact of exogenous variables and data types on accuracy, and creating an updated framework for forecasting SPVPO

and WTPO. These focused efforts can help overcome existing limitations and tap into new technological advancements, ultimately enhancing the accuracy, reliability, and efficiency of renewable energy forecasting using machine learning techniques.

Abbreviations

ACE	Adaptive calibration error
AD	Air density
AI	Artificial intelligence
ANN	Artificial neural network
ANOVA	Analysis of variance
AP	Air pressure
AT	Ambient temperature
BCT	Back-cell temperature
CC	Cloud cover
CLOT	Cloud optical thickness
CNN	Convolutional neural networks
CRPS	Continuous ranked probability score
EM	Ensemble model
EMD	Empirical mode decomposition
EMS	Energy management system
FCM	Fuzzy C-means
G	Geopotential
GA	Genetic algorithm
GBM	Gradient boosting machines
HBOS	Histogram-based outlier score
HF	Heat flux
HM	Hybrid models
IF	Isolation forest
IoT	Internet of things
L	Losses
LSTM	Long short-term memory
MAPE	Mean absolute percentage error
ML	Machine learning
MLP	Multilayer perception
MRE	Mean relative error
MSE	Mean absolute error
NWP	Numerical weather prediction
PCA	Principal component analysis
PHM	Physical hybrid model
PICP	Prediction interval coverage percentage
PPO	Prior power output
RF	Random forests
RH	Relative humidity
RMSE	Root mean square error
RMSPE	Root mean squared percentage error
RNN	Recurrent neural networks
ROPES	Reproducible, operational, probabilistic/physically based, ensemble, and skill
SA	Solar angle
SARIMAX	Seasonal autoregressive integrated moving average with exogenous variables
SD	Sunshine duration
SPVPO	Solar photovoltaic power output
SVM	Support vector machines
TA	Tilt angle
Tb	Turbulence
TP	Precipitation
TPA	Position and angles
TSp	Turbine speed
TT	Temperature
WB	Wind bearing
WD	Wind direction
WS	Wind speed
WT	Weather type
WTPO	Wind turbine power output
XGBoost	Extreme gradient boosting

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Author contributions

The first author contributed to the conception, study design, execution, acquisition of data, analysis, interpretation and drafting of the manuscript. The second author supervised the study, critically reviewed the works, and corrected the final manuscript.

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